

RASCAL versions: Detailed capability comparison

Desktop Python, HTML WebGL, ImageJ/Fiji plugin, and Android

Scope. This document compares the practical capabilities of the four supplied RASCAL implementations. The Desktop Python version is treated as the reference implementation; the other versions are assessed as lighter or more specialized versions for web demonstration, ImageJ/Fiji workflows, or field use on Android devices.

Version	Best use case
Desktop Python	Complete reference version for serious processing, reproducible work, publication figures, demanding images, and 3D textured models.
HTML WebGL	Very convenient zero-install version for demonstrations, quick exploration, teaching, and sharing. It runs locally in the browser but remains functionally lighter.
ImageJ/Fiji plugin	Best suited for users already working in the ImageJ/Fiji ecosystem. It gives an interactive RASCAL preview inside a familiar scientific imaging workflow, which can be easily compared to DStretch® outputs.
Android	Most appropriate version for field exploration on a phone or tablet, with gallery/camera workflows and touch interaction. It is intentionally simplified.

1. Platform, deployment and intended use

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
Platform	Windows, macOS, Linux	Any recent browser with WebGL2	ImageJ/Fiji on Windows, macOS, or Linux	Android 8 or later
Installation model	Standalone application or Python environment	Single HTML file; no installer	JAR file placed in the ImageJ/Fiji plugins folder	APK installation
Main target user	Advanced users, publication work, difficult cases	Demonstration, teaching, fast sharing	ImageJ/Fiji users and lab workflows	Field users on phone/tablet
Privacy / processing location	Local processing	Local browser processing; no upload in the supplied interface	Local processing inside ImageJ/Fiji	Local processing on the device
Dependency risk	Qt/OpenGL/VisPy	Browser and WebGL2 support are critical	Java/ImageJ/Fiji/JOGL compatibility matters	Android/OpenGL ES and device memory matter

2. Input, image formats and large files

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
Built-in image loading	Yes	Yes	No: the image must first be opened in ImageJ/Fiji	Yes: both gallery and camera workflows
JPEG / PNG / BMP	Yes	Yes, through browser support	Yes if opened by ImageJ/Fiji	Yes, depending on Android support
TIFF	Yes, including high-bit-depth TIFF handling	No / very limited; TIFF is not a target format	Possible if ImageJ/Fiji opens it	No
Camera RAW / DNG	Yes	No	Not directly; use ImageJ/Fiji import/conversion first	No
Very large images	Best option, but memory can still become limiting	Limited; the supplied code reduces large images	Depends on ImageJ/Fiji memory settings	Limited; large images may be downsampled

3. Core RASCAL processing

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
sRGB to linear RGB conversion	Yes	Yes	Yes	Yes
ZCA whitening	Yes	Yes	Yes	Yes
Interactive RGB-space rotation	Yes	Yes	Yes	Yes
X/Y/Z sliders	Yes	Yes	Yes	Yes
Mouse / touch rotation	Yes; mouse interaction plus sliders	Yes; mouse/touch interaction plus sliders	Yes; preview interaction plus sliders	Yes; touch interaction plus sliders
Color transformed view	Yes	Yes	Yes	Yes
R/G/B grayscale channel views	Yes	Yes	Yes	Yes
Push factor	Yes	Yes	Yes	Yes
Variance restore	Yes	Yes	Yes	Yes
ICA orientation suggestion	Yes	Yes	Yes	Yes

4. Interaction and exploration

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
Basic / Expert modes	Yes	No	No	No
Random rotation explorer	Yes	Yes	Yes	Yes
Continuous auto-rotation	Yes	No	No	No
Split original/processed view	Yes	Yes	Yes	Yes
Polarity inversion / Pol. +/-	Yes	Yes	Yes	No
Temporary presets	Yes, up to 15	Yes, up to 5	Yes, up to 10	Yes, up to 5
Persistent presets after closing	No	No	No	No
Comfort of detailed exploration	Highest: full desktop layout, panels, 3D cloud and export tools	Good for quick exploration; interface is responsive	Good if the user already works in ImageJ/Fiji	Good on tablets/phones, but simplified

5. ROI, advanced processing and 3D functions

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
ROI painting	Yes	Yes	Yes	Yes
Comfort of ROI use	Very good, with dedicated tools and brush controls	Good	Good	Good
Radial Push / Fibonacci radial normalization	Yes	No	No	No
Use of ROI for Radial Push reference	Yes	No	No	No
3D RGB or whitened color cloud	Yes	No	No	No
3D textured OBJ/PLY model loading	Yes	No	No	No
RASCAL processing on 3D textured models	Yes	No	No	No
Image quality report	Yes	No	No	No

RASCAL capability comparison

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
Image modification tools inside RASCAL	Yes: crop, contrast, HSL/channel adjustments and resizing	No	No, but ImageJ can preprocess externally	Very limited; mainly camera and image loading workflows

6. Export, reproducibility and publication workflows

Feature / capability	Desktop Python	HTML WebGL	ImageJ/Fiji plugin	Android
Export processed image	Yes	Yes, PNG export	Yes, as processed image/JPEG workflow	Yes, JPEG saved to the device
Export original + processed comparison	Yes, including split/export options	Limited to processed view/workflow	Limited to processed view/workflow	Limited to processed view/workflow
Export separate channel views	Yes	No	No	No
Export 3D color cloud	Yes	No	No	No
Export processed 3D model texture	Yes	No	No	No
Session save/load	Yes, .rasc session files	No	No	No
Reproducibility of processing parameters	High: session can store key parameters and state	Low to medium: image export only	Low to medium: image export only	Low to medium: image export only
Publication-grade workflow	Best suited	Mostly demonstration and exploration	Good if embedded in a documented ImageJ workflow	Mostly field documentation and exploration

7. Overall assessment

Version	Assessment
Desktop Python	Most complete and most scientifically reproducible version. It is the best choice for difficult images, high-bit-depth files, systematic processing, figure preparation and 3D textured models. Its main weakness is memory use on very large images.
HTML WebGL	Excellent for diffusion: easy to share, no installer, local processing, and responsive interface. Its main limits are the absence of Radial Push, sessions, 3D, RAW/TIFF support, and robust large-file handling.
ImageJ/Fiji plugin	Useful when the user already relies on ImageJ/Fiji and for comparison with DStretch. It benefits from ImageJ import and preprocessing tools, while providing a RASCAL preview. It remains much lighter than the desktop version: no Radial Push, no sessions, no 3D cloud, no 3D textured model workflow.
Android	Best field version. It is convenient for quick visual checks, gallery/camera use, tactile ROI and on-site exploration. It is intentionally simplified: no RAW/TIFF, no Radial Push, no sessions, no 3D, and less control over very large images.

Recommended hierarchy. Desktop = complete and reproducible reference version; HTML WebGL = demonstration and rapid dissemination version; ImageJ/Fiji = integration into an established scientific imaging workflow, often including DStretch processing, especially for rock art specialists; Android = mobile field exploration version.